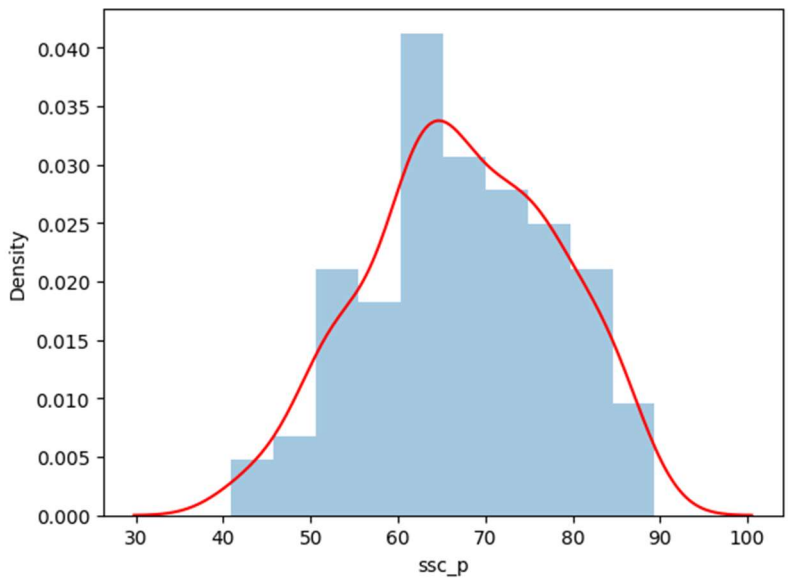
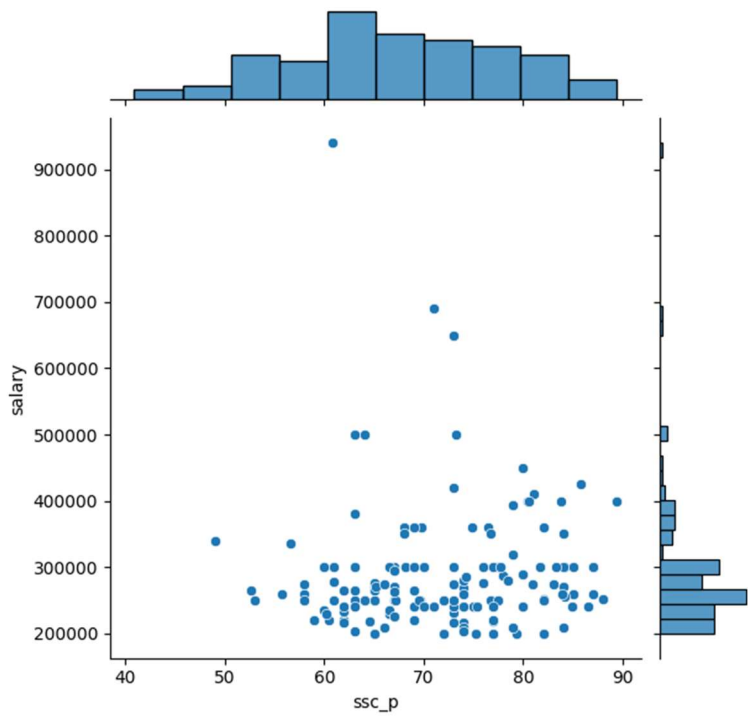
1. Distplot - Histogram (SSC marks)



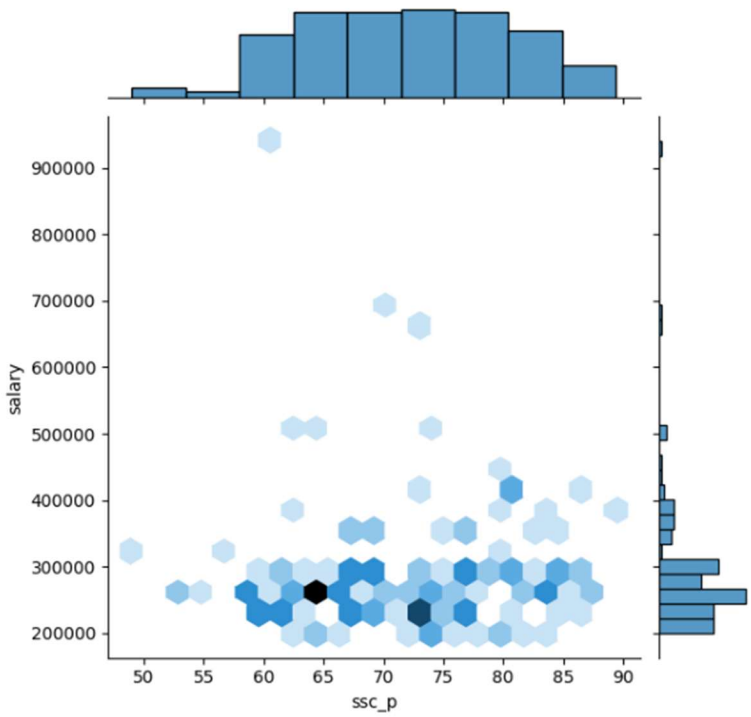
- The SSC percentage is mostly concentrated between 60 - 80, which shows most students scored in the average to high range.

2. Plot (Scatter) - SSC vs Salary



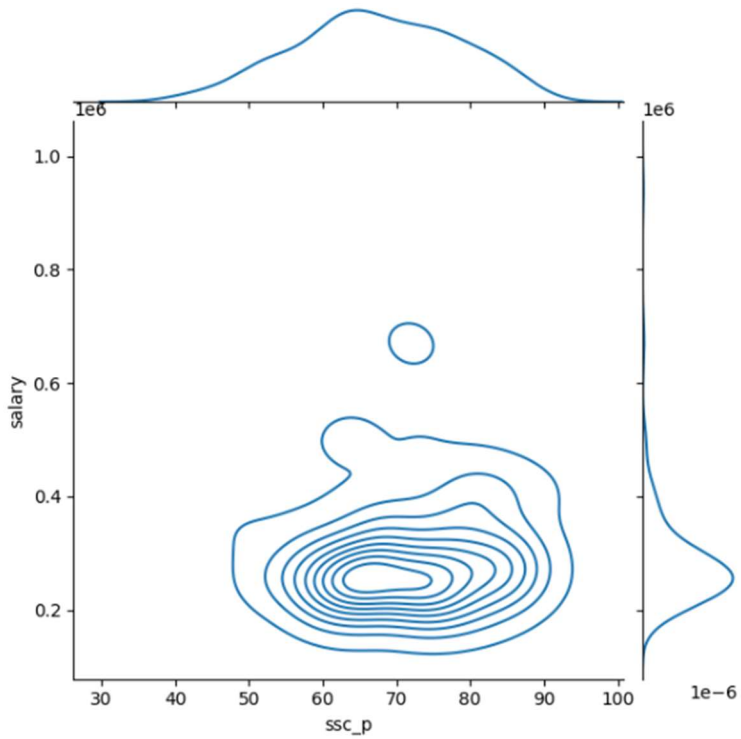
- This plot shows the relationship between SSC marks and salary.
- Most salaries are around 200k - 300k.
- The points are scattered, which indicates there is only a weak relationship between SSC marks and salary.

3. Joint Plot - Hexbin



- The hexbin plot shows where most students are concentrated.
- The darkest region indicates many students have SSC marks around 60 -75 and salaries around 200k - 300k.

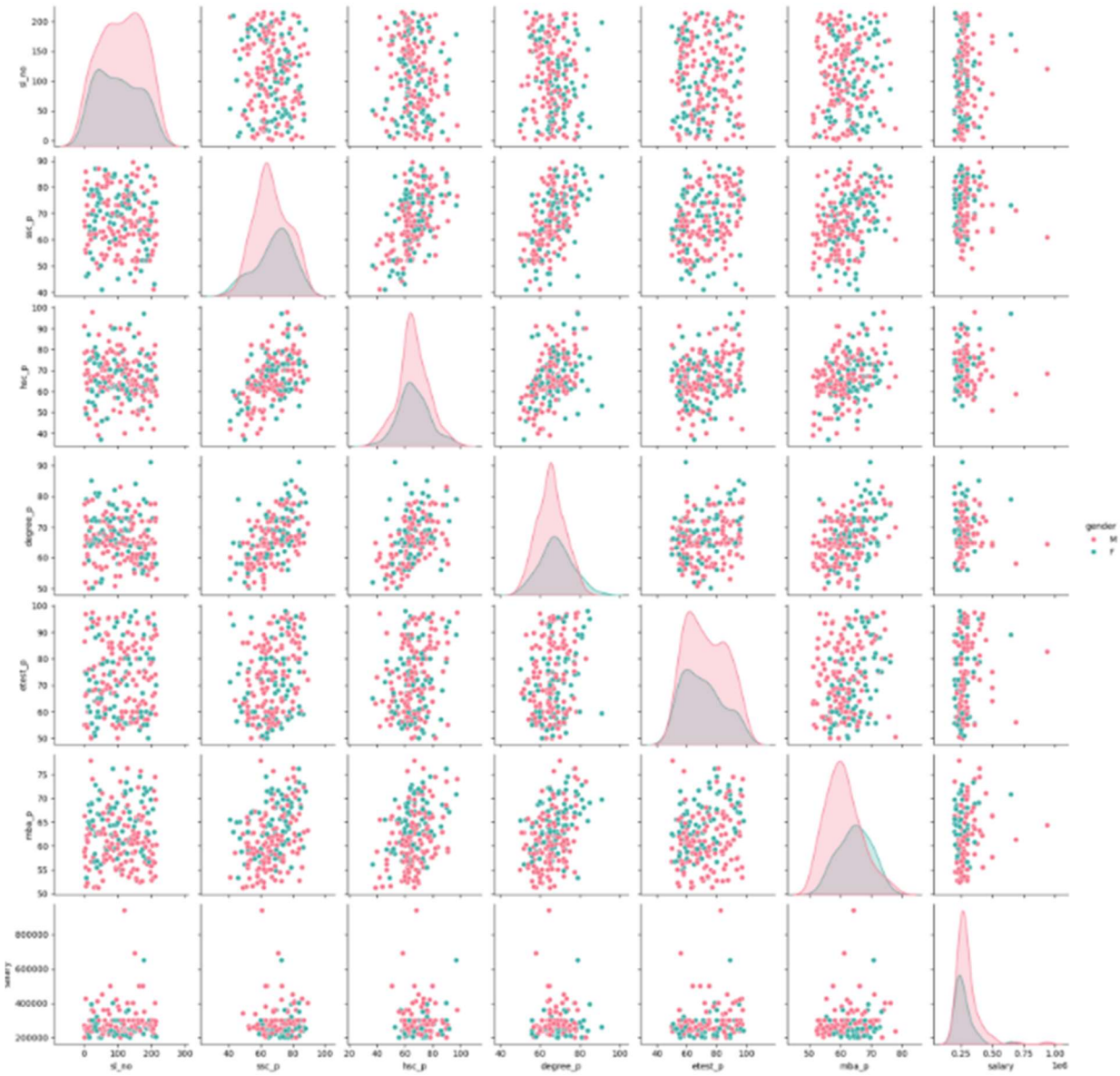
4. Joint Plot - KDE (Density Contours)



- The contour lines represent areas where data points are most concentrated.

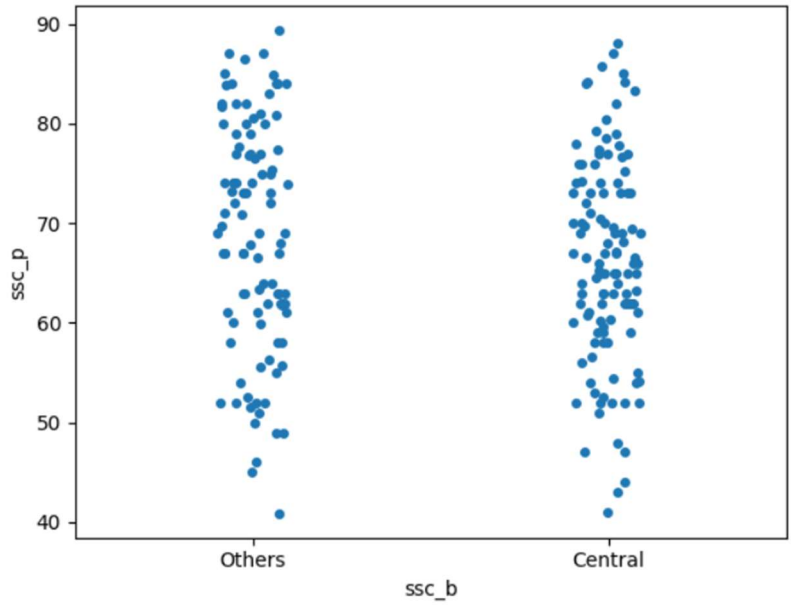
- The KDE contour plot shows the density of data points, indicating that most students have SSC marks around 60 - 75 and salaries around 200k - 300k.

5. Pair Plot - All Variables



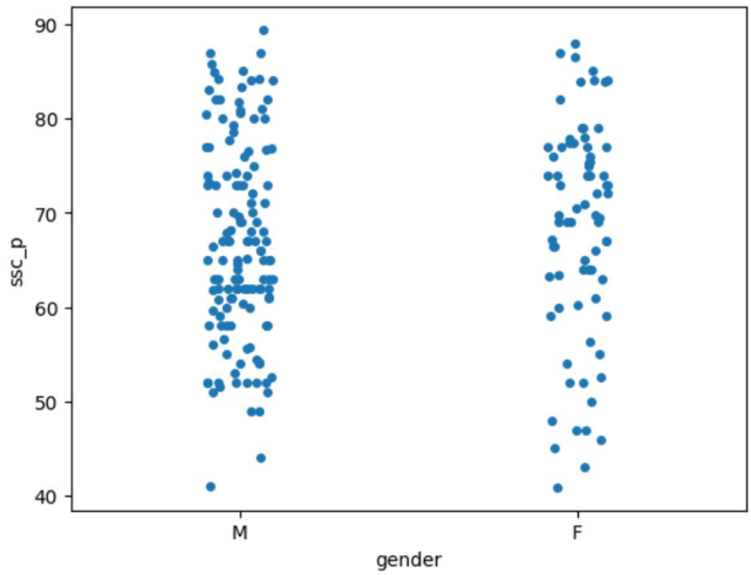
- The pairplot shows relationships between all numerical variables.
- It helps visualize how different academic scores and salary are related to each other.
- Academic marks show some positive relationships.
- Salary values are mostly concentrated in lower ranges.

6. Strip Plot - SSC Board vs Marks



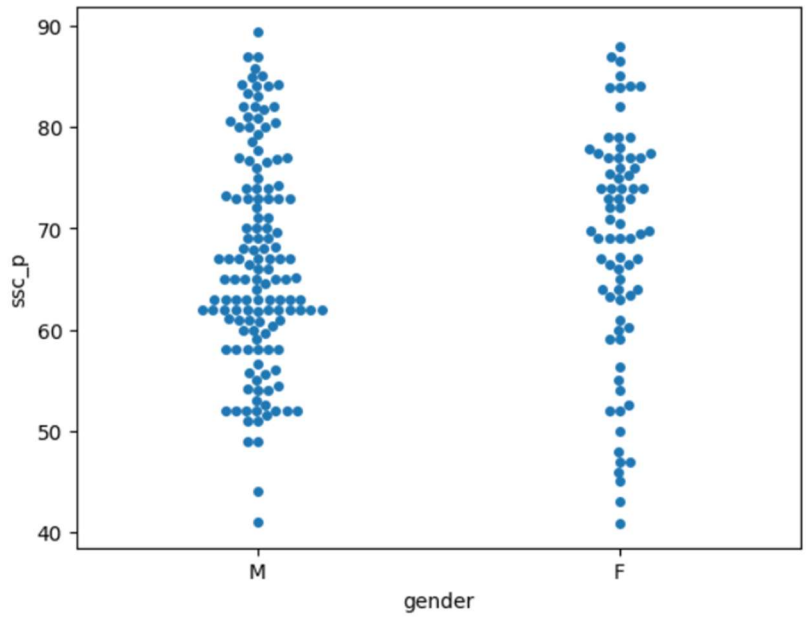
- The strip plot shows individual SSC marks across different boards, which means that both (others & central) boards have similar spread of marks.

7. Strip Plot - Gender vs SSC Marks



- The SSC marks distribution for male and female students appears similar, with no large difference between genders.

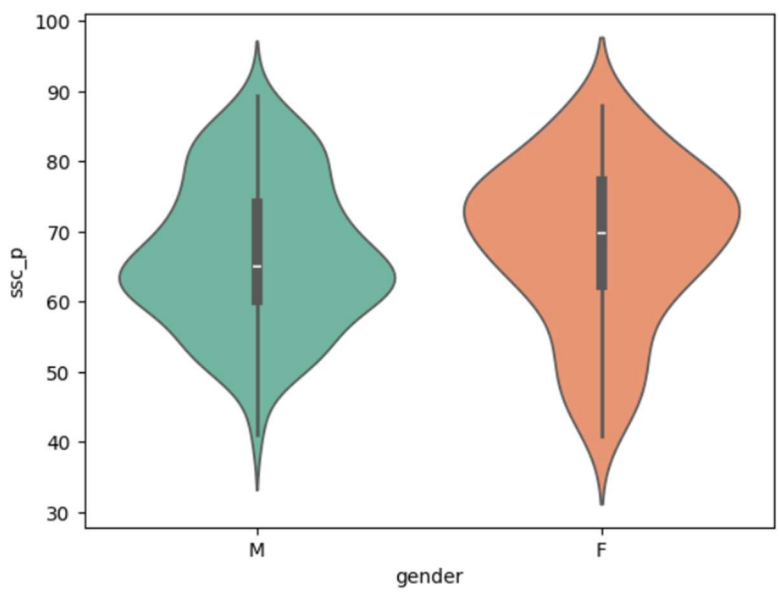
8. Swarm Plot - Gender vs SSC Marks



- Same as strip plot but points don't overlap, so the distribution is clearer.

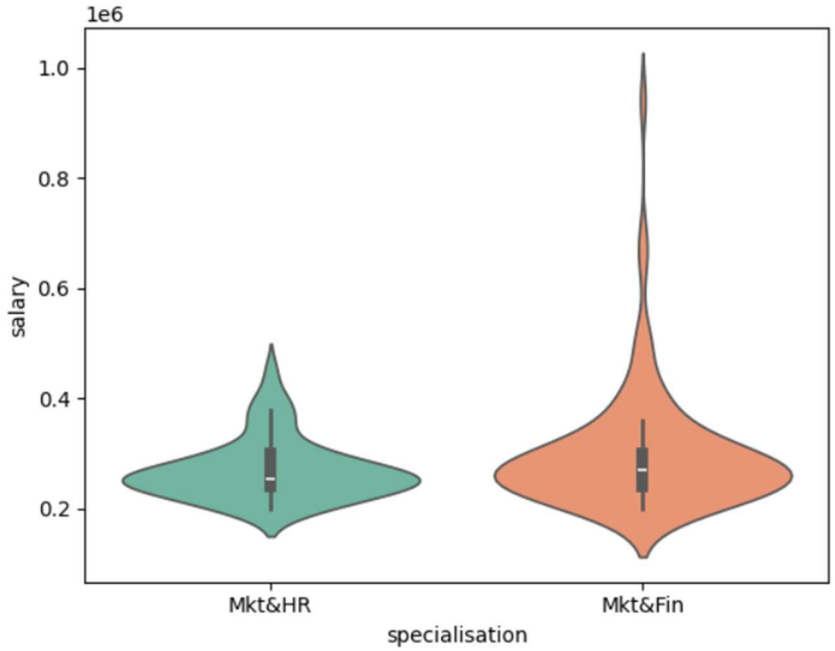
- The swarm plot shows the distribution of SSC marks for each gender without overlapping points, making it easier to observe the spread of scores.

9. Violin Plot - Gender vs SSC Marks



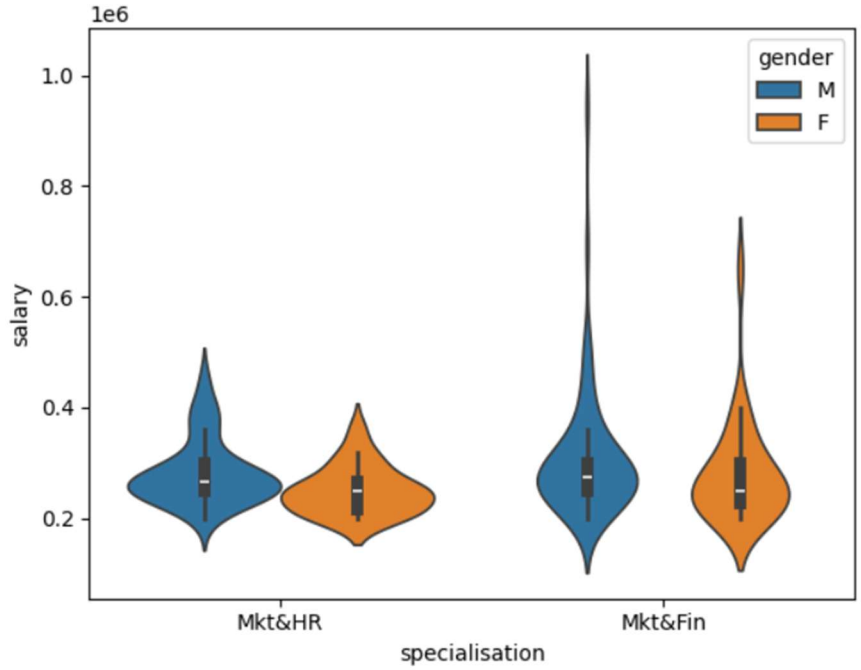
- The violin plot Shows distribution + density.
- Distribution and density of SSC marks show similar score distributions for male and female students.
- Female marks appear slightly higher concentrated around 70 - 75.

10. Violin Plot - Specialisation vs Salary



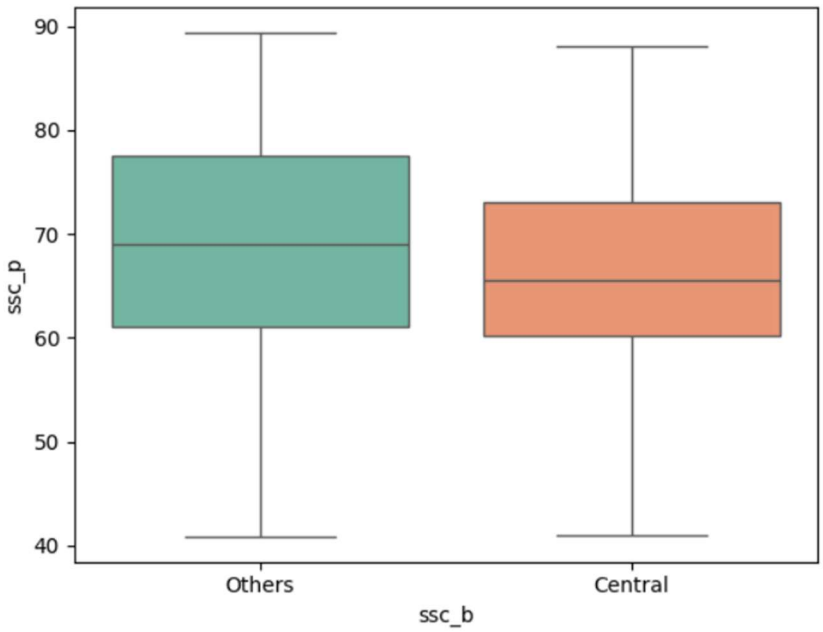
- The violin plot shows salary distribution by specialization.
- Mkt&Fin has higher salary spread and some higher salary values than Mkt&HR students.

11. Violin Plot - Specialisation + Gender vs Salary



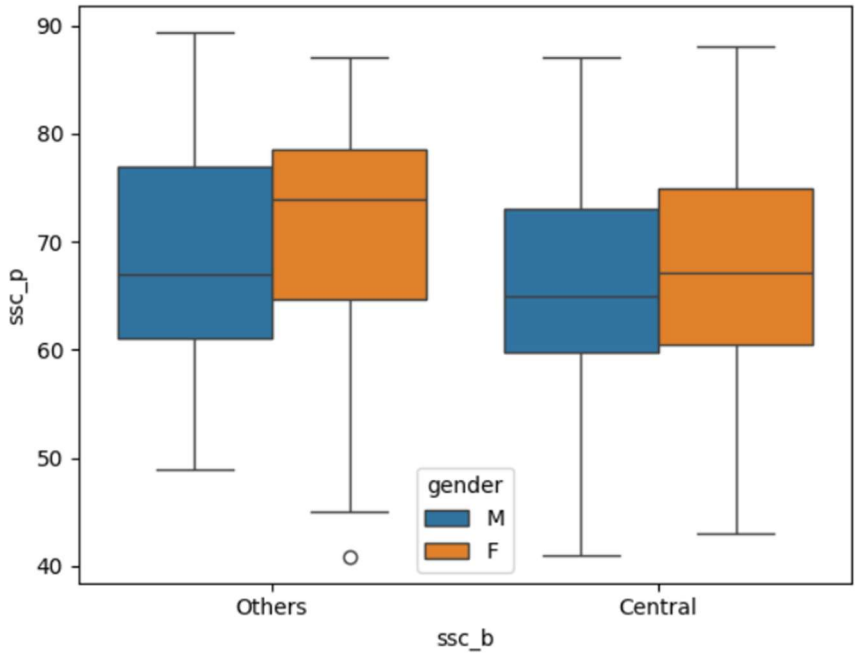
- This plot compares salary distribution across specialization and gender.
- We can see that the salary patterns are similar across genders but vary slightly by specialization.
- Male students of Mkt&Fin has higher salary spread and some higher salary values than female students in that department.

12. Box Plot - SSC Board vs SSC Marks



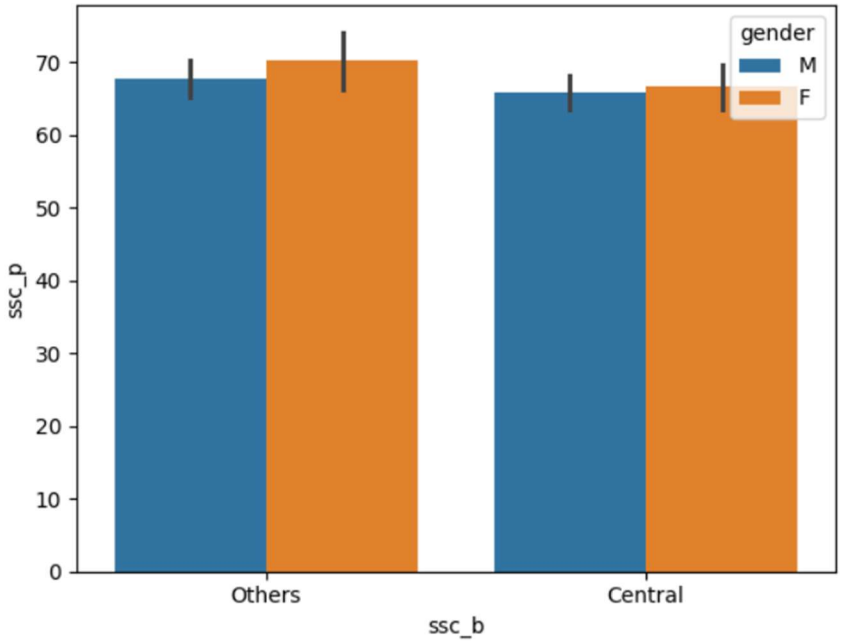
- The boxplot shows the distribution of SSC marks by board.
- The median SSC marks of students from other boards is slightly higher than central board students.
- The overall range of SSC marks is similar for both boards, roughly from about 40 to around 90 marks.
- Most students in both boards have SSC marks roughly between 60 and 75.

13. Boxplot - SSC Board vs SSC Marks by Gender



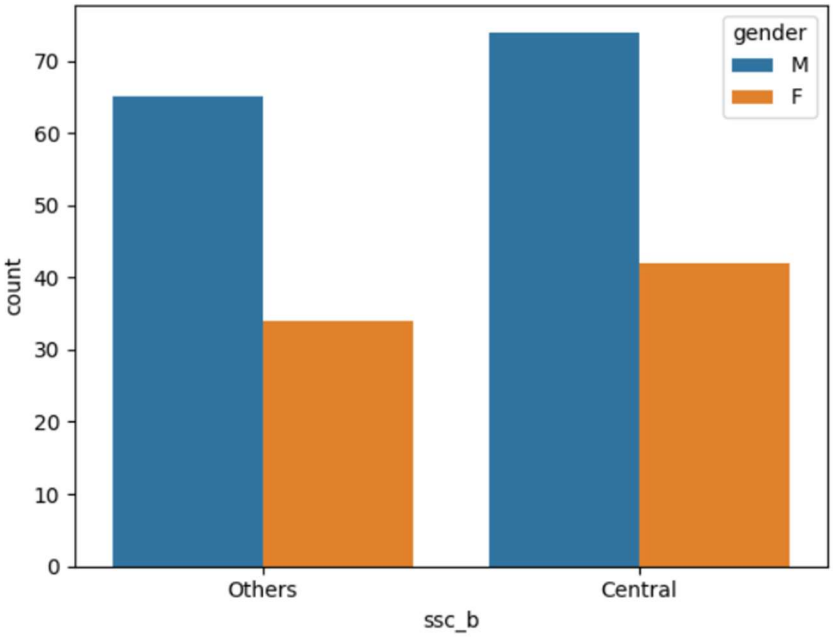
- Female students have slightly higher median SSC marks than male students in both boards.
- SSC marks for both genders are mostly distributed between 60 and 75 marks.
- There is one lower outlier for female students in the others board.

14. Bar Plot - Average SSC Marks by Board and Gender



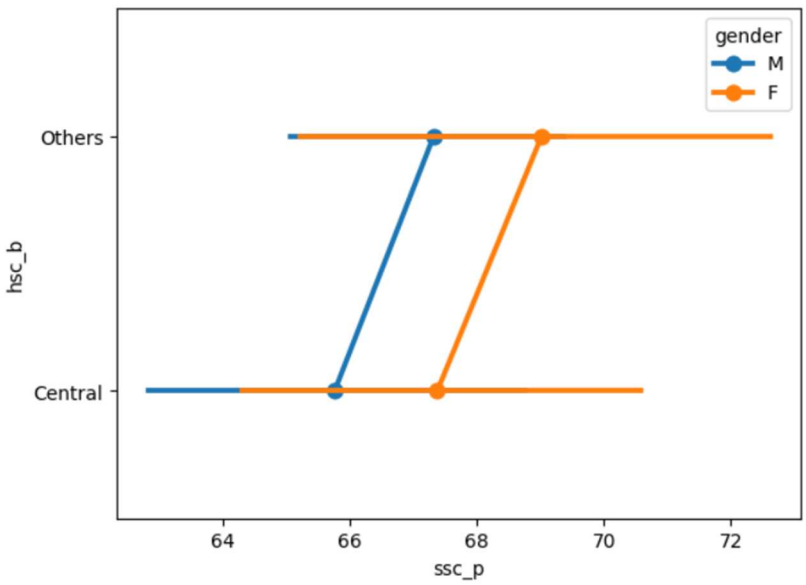
- Female students have higher average SSC marks than male students in both boards.
- Students from Other boards show slightly higher average marks than Central board students.
- The difference between groups is small and not very large.

15. Count Plot - Number of Students by SSC Board and Gender



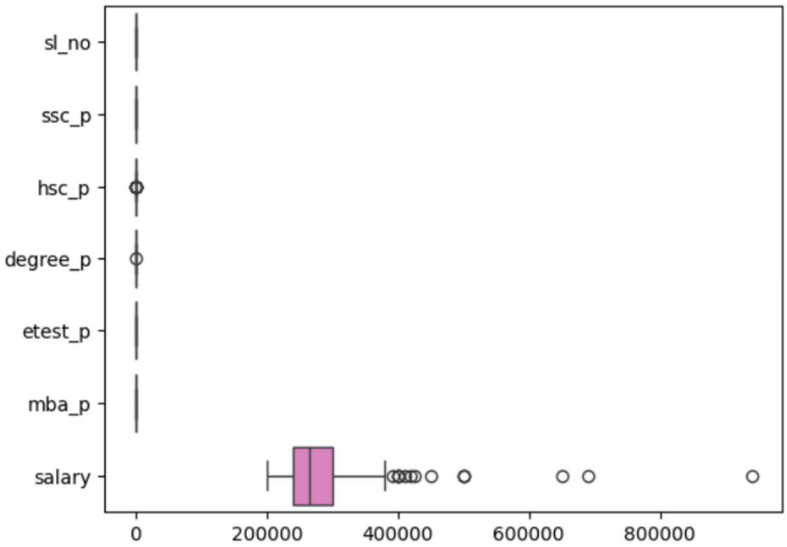
- There are more male students than female students in both boards.
- The Central board has more students overall than other boards.

16. Point Plot - SSC Marks vs HSC Board by Gender



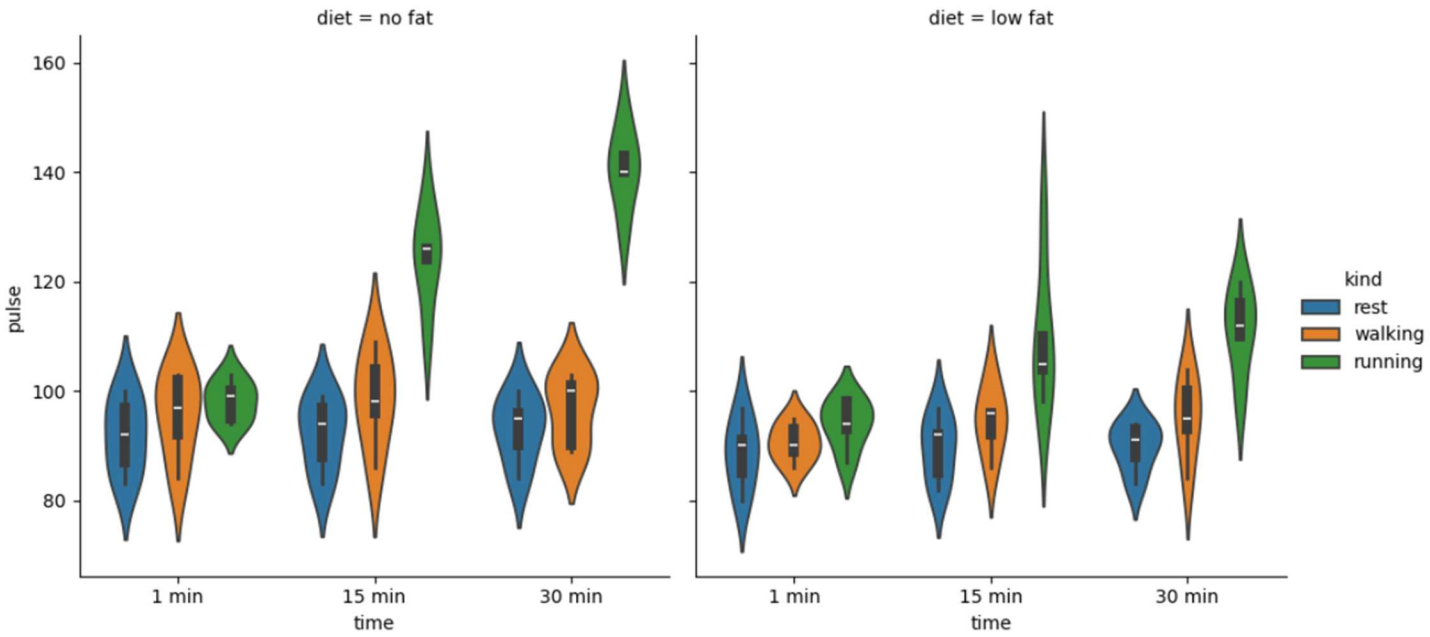
- Female students show slightly higher SSC marks compared to male students.
- SSC marks are slightly higher for students from Other boards than Central board.
- The difference between the groups is small but consistent.

17. Horizontal Boxplot - All Numerical Variables



- Salary shows several outliers, indicating some students received very high salary packages.
- Most salary values are concentrated around 200000 - 300000 range.
- Academic marks (SSC, HSC, Degree, Entrance test, MBA) are distributed within a smaller range compared to salary.

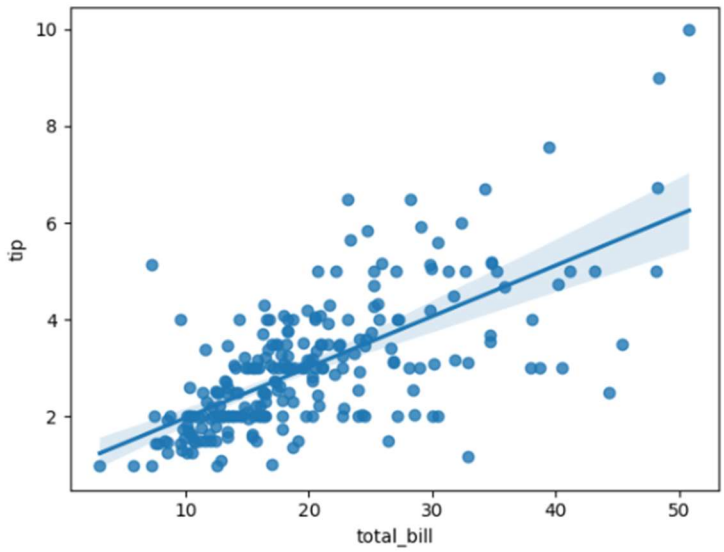
18. Violin plot - Pulse rate distribution during different exercises and diet types.

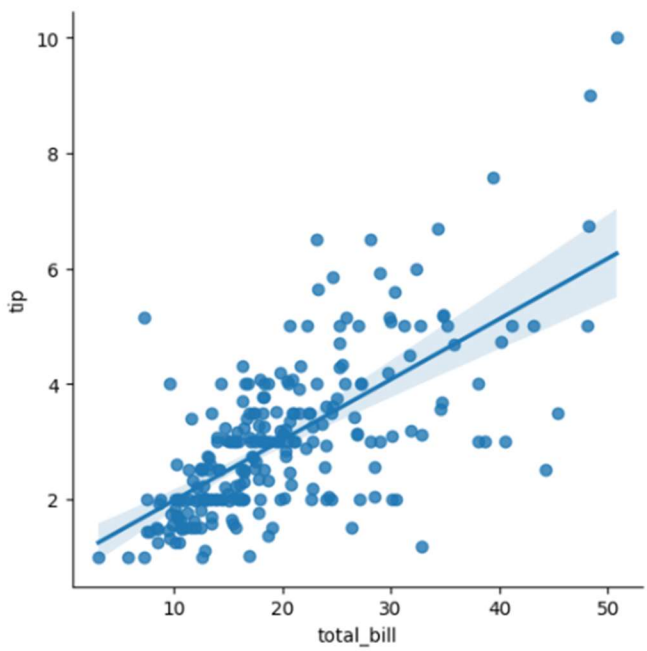


- This violin plot shows the pulse rate of people at different exercise times (1 min, 15 min, 30 min) for two diet groups (no fat diet and low-fat diet) and three activity types (rest, walking, running).
- Pulse rate during rest is the lowest for both diet groups at all time intervals.
- Pulse rate during walking is slightly higher than rest, but still lower compared to running.
- Pulse rate during running is the highest, especially after 15 minutes and 30 minutes.
- As exercise time increases (1 → 15 → 30 minutes), the pulse rate generally increases, especially for running.
- In the no fat diet group, running shows a much higher pulse rate at 30 minutes compared to the low-fat diet group.
- In the low-fat diet group, pulse rates are slightly lower compared to the no fat diet group for similar activities.
- The spread of the violin shape shows that pulse values vary more during running than rest or walking.

19. Regression Plot (Scatter Plot with Regression Line) - Relationship between Total Bill Amount and Tip Given

```
[31]: import pandas as pd
      import seaborn as sb
      from matplotlib import pyplot as plt
      df = sb.load_dataset('tips')
      sb.regplot(x = "total_bill", y = "tip", data = df)
      sb.lmplot(x = "total_bill", y = "tip", data = df)
      plt.show()
```





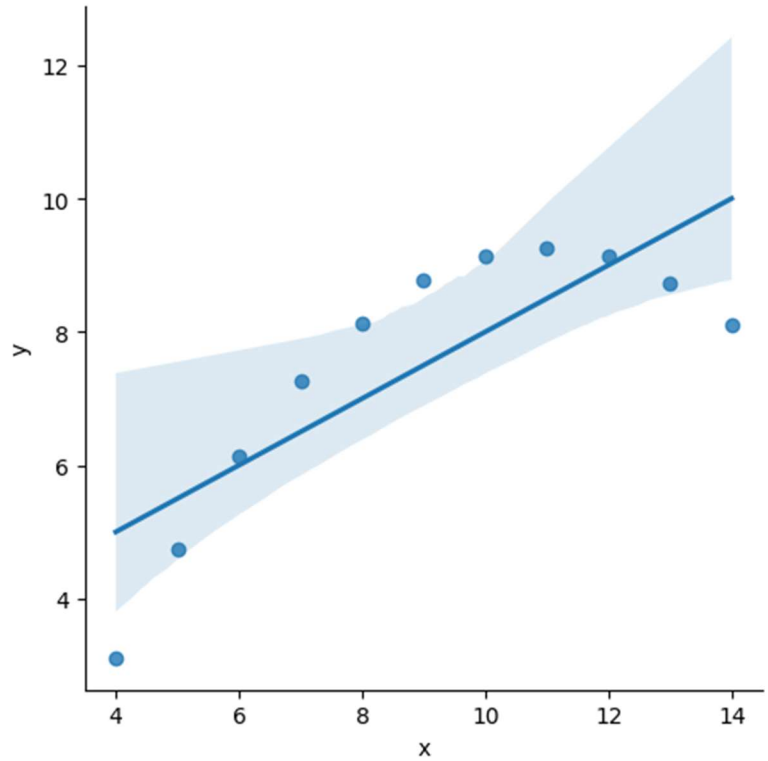
- There is a positive relationship between total bill and tip. When the bill amount increases, the tip amount also increase.
- Most data points are concentrated around bill values between 10 and 30 and tips between 2 and 4.
- The regression line slopes upward, which means that customers generally give higher tips for higher bills.
- Some points are scattered away from the line, which means not all customers tip the same percentage, and tipping behavior varies.

regplot → Used to create Simple regression plot

lmplot → Used to create regression plot with grouping (hue, col, etc.)

20. Linear Regression Plot - Relationship between X and Y values in Anscombe Dataset (Dataset II)

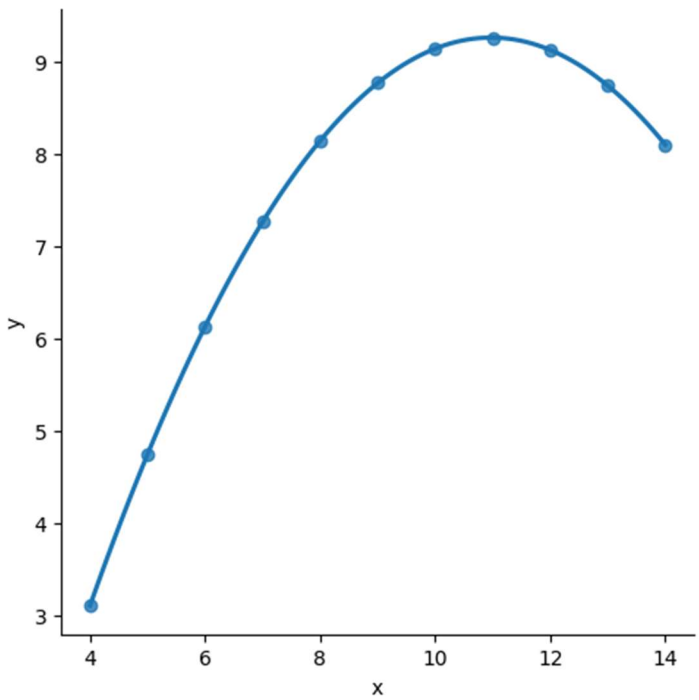
```
import pandas as pd
import seaborn as sb
from matplotlib import pyplot as plt
df = sb.load_dataset('anscombe')
sb.lmplot(x = "x", y = "y", data = df.query("dataset == 'II'"))
plt.show()
```



- The plot shows a positive linear relationship between X and Y. As X increases, Y also increases.
- Most data points are close to the regression line, which shows a strong linear trend.
- There are a few points slightly away from the line, but overall the data follows the general linear pattern.

21. Polynomial Regression Plot (Non-linear relationship between X and Y values in Anscombe Dataset)

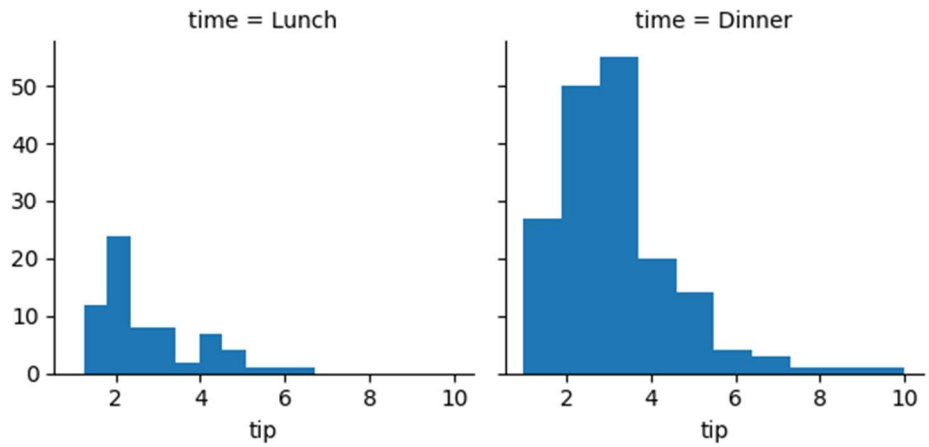
```
import pandas as pd
import seaborn as sb
from matplotlib import pyplot as plt
df = sb.load_dataset('anscombe')
sb.lmplot(x = "x", y = "y", data = df.query("dataset == 'II'"),order = 2)
plt.show()
```



- The curve shows a non-linear relationship between X and Y.
- Y values increase as X increases up to around 10 - 11 and then start to decrease slightly.
- The points follow a curved pattern, which tells that the relationship is quadratic rather than linear.
- Most points lie close to the curve, showing that the polynomial regression fits the data well.

22. FacetGrid Histogram Plot - Distribution of Tip Amounts by Meal Time (Lunch vs Dinner)

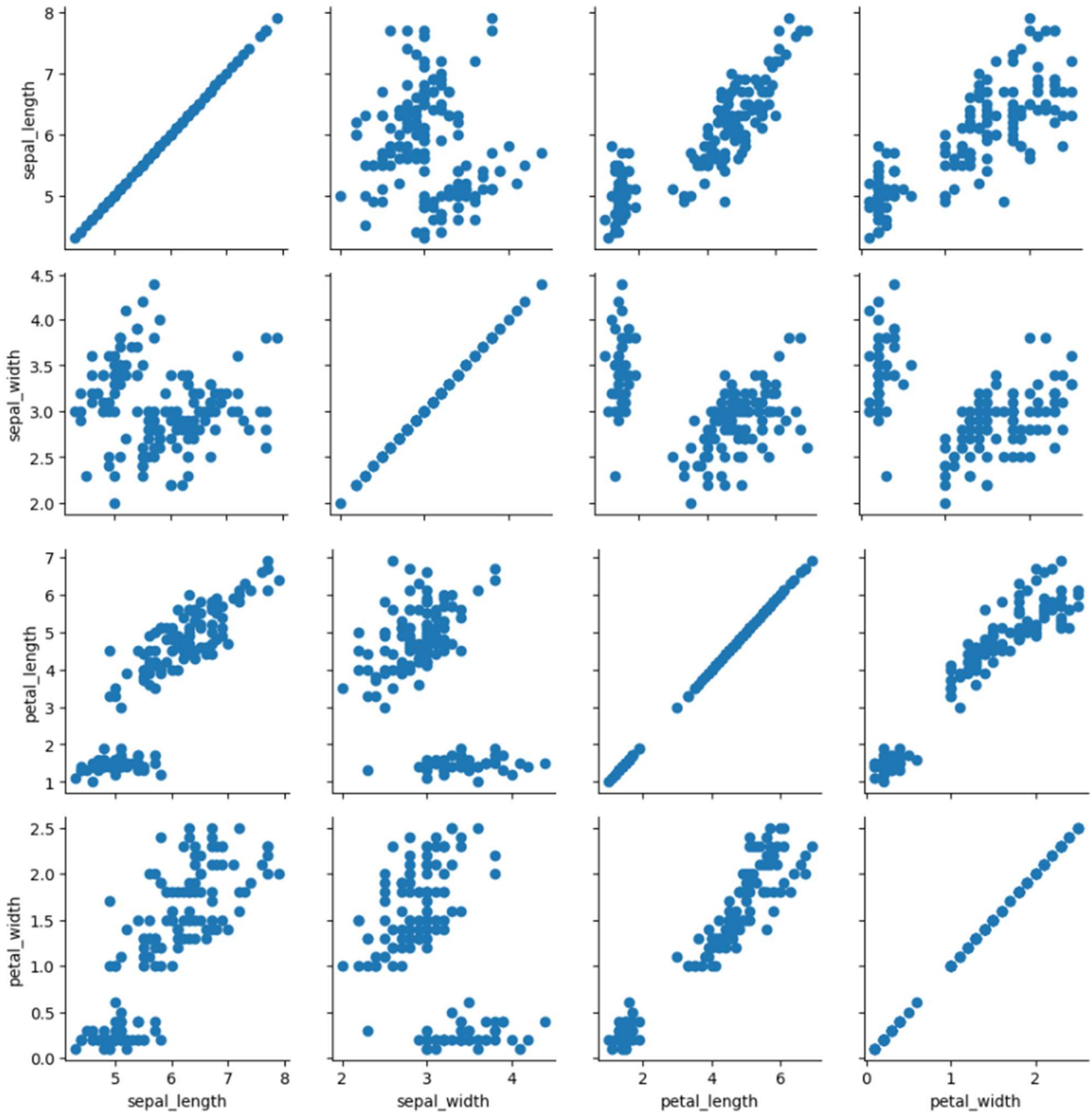
```
[34]: import pandas as pd
      import seaborn as sb
      from matplotlib import pyplot as plt
      df = sb.load_dataset('tips')
      g = sb.FacetGrid(df, col = "time")
      g.map(plt.hist, "tip")
      plt.show()
```



- Most tip values during lunch are between 1 and 3.
- Dinner has more observations and higher tip values compared to lunch.
- The distribution of dinner tips is wider, meaning tips vary more during dinner.
- Very high tips are rare in both lunch and dinner.

23. **PairGrid Scatter Plot - Relationship between sepal and petal measurements in the Iris dataset**

```
[35]: import pandas as pd
import seaborn as sb
from matplotlib import pyplot as plt
df = sb.load_dataset('iris')
g = sb.PairGrid(df)
g.map(plt.scatter);
plt.show()
```



- The plot shows the relationship between all numerical variables such as sepal length, sepal width, petal length, and petal width.
- Petal length and petal width show a strong positive relationship, meaning when petal length increases, petal width also increases.
- Sepal length also increases with petal length and petal width, showing a moderate positive relationship.
- Sepal width does not show a strong relationship with the other variables, as the points are more scattered.
- The diagonal plots show the same variable compared with itself, which results in a straight line.